Unit 5 Summary: Study Design

Name _____

Types of Studies

Type	What happens	Example
Sample survey	Ask a sample questions.	Ask 100 students how many hours they sleep.
Observational study	Watch or measure. Do **not** change anything.	Compare the grades of students who do and do not play sports.
Experiment	**Do something** to the subjects (a treatment) and measure what happens.	Give half the students a 10-minute break before a test. Compare scores.

Choosing a Sample

Method	How it works	Example: 60 of 1,200 students
Simple random sample (SRS)	Every group of the same size has the same chance of being chosen.	A random number generator picks 60 student numbers.
Stratified	Split the population into groups that are alike (**strata**). Take a random sample from **each** group.	Randomly pick 15 students from each grade.
Cluster	Split the population into groups (**clusters**). Randomly pick some **whole** groups.	Randomly pick 3 homerooms. Ask everyone in them.
Systematic	Pick a random start. Then take every *k*th person.	Start at a random student from 1 to 20. Take every 20th student.
Convenience (bad)	Take whoever is easy to reach. No chance is used.	Ask the first 60 students who walk in the door.

Stratified: a **few** from **every** group. **Cluster: everyone** from a **few** groups.

Bias

Type of bias	What happens	Example
Voluntary response	People choose if they answer. People with strong feelings answer most.	A radio call-in poll.
Convenience	The sample is whoever is easy to reach.	Asking only your friends.
Undercoverage	Some groups are left out of the sample.	Calling only home phones.
Nonresponse	People who were chosen do not answer.	Half of the mailed surveys never come back.
Response	The question or the setting pushes people to answer a certain way.	“Don’t you agree that our great principal deserves a raise?”

Experiment Designs

Design	How it works	Example
Completely randomized	Chance alone puts each subject in a treatment group.	40 volunteers: chance picks 20 for the new pill and 20 for the placebo.
Randomized block	Split the subjects into groups that are alike (**blocks**). Then chance assigns the treatments inside each block.	40 gym members. The blocks are beginners and experienced members. Chance picks who gets each workout plan inside each block.
Matched pairs	Each subject gets **both** treatments, in a random order. Or pair up very similar subjects, and chance picks who gets which.	Each runner runs once in each shoe. A coin flip decides which shoe goes first.

A good experiment has: a control group to compare to, random assignment to make the groups alike, and replication (many subjects).

Blocks and strata: both split people into groups that are alike, then use chance inside each group. In a stratified sample, chance picks who is in the sample. In a randomized block design, chance picks which treatment each subject gets.

What Can a Study Prove?

	Random assignment	No random assignment
Random sample	Cause: yes. Generalize: yes.	Cause: no. Generalize: yes.
Not a random sample	Cause: yes, for these subjects. Generalize: no.	Cause: no. Generalize: no.

Statistically significant: if chance gives a result this big less than about 5% of the time, the result is probably real.